Maijunxian Wang

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Education

University of California, Berkeley Sep 2025 - Spring 2027 (expected)
B.A. in Cognitive Science GPA: 3.965/4.0

University of California, Davis Fall 2023 - Summer 2025
Undergraduate coursework in Cognitive Science; transferred to UC Berkeley GPA: 3.973/4.0

Research Interests

Visual reasoning; video generation and world models; cognitively grounded AI.

I study vision as a reasoning medium distinct from language, using cognitively grounded tasks and video-model training to develop and evaluate general visual reasoning capabilities.

Honors and Awards

Sayan Gul Award - Best Undergraduate Student Paper CogSci 2026
Cognitive Science Society, for *Egocentric Bias in Vision-Language Models*.

Publications

* Equal contribution. Long author lists are abbreviated with ellipses; author order is preserved.

Conference Papers

A Very Big Video Reasoning Suite
Maijunxian Wang*, Ruisi Wang*, Juyi Lin*, Ran Ji*, . . . , Zhongang Cai, Hokin Deng.
ICML, 2026.

Egocentric Bias in Vision-Language Models
Maijunxian Wang*, Yijiang Li*, Bingyang Wang, Tianwei Zhao, Ran Ji, Qingying Gao, Emmy Liu, Hokin Deng, Dezhi Luo.
CogSci, 2026. Sayan Gul Award.

Assessing Perceptual Metacognition in Vision-Language Models
Ran Ji*, **Maijunxian Wang***, Qingying Gao, Yijiang Li, Hokin Deng, Dezhi Luo.
CogSci, 2026.

Demystifying Video Reasoning
Ruisi Wang, Zhongang Cai, Fanyi Pu, . . . , **Maijunxian Wang**, . . . , Ziwei Liu, Lei Yang.
ECCV, 2026.

Vision Language Models Cannot Reason About Physical Transformation
Dezhi Luo*, Yijiang Li*, **Maijunxian Wang**, Tianwei Zhao, Bingyang Wang, Siheng Wang, Pinyuan Feng, Pooyan Rahmanzadehgervi, Ziqiao Ma, Hokin Deng.
ICML, 2026.

Preprints

VBVR-Pro: A Scalable and Verifiable Suite for Native Visual Reasoning
Junxiang Xu*, Ruisi Wang*, Fanyi Pu*, **Maijunxian Wang***, Ran Ji*, Tongxi Zhou*, . . . , Hokin Deng, Zhongang Cai.
Preprint, 2026.

Training Object Permanence in World Models
Haotian Zhang, Fengyuan Yu, Dezhi Luo, . . . , **Maijunxian Wang**, . . . , Yilun Du, Hokin Deng.
Preprint, 2026.

Workshop Papers

Increasing Computation Resolves Conflicts in Vision Language Models
Bingyang Wang*, Yijiang Li*, Yitong Qiao, **Maijunxian Wang**, Tianwei Zhao, Yucheng Sun, Binyue Deng, Hokin Deng, Nuno Vasconcelos, Dezhi Luo.
MMRAgI Workshop at CVPR, 2026.

Probing Perceptual Constancy in Large Vision-Language Models
Haoran Sun, Bingyang Wang, Suyang Yu, . . . , **Maijunxian Wang**, Dezhi Luo, Hokin Deng.
ES-Reasoning Workshop at ICLR, 2026.

Research Experience

Hippocampal May 2026 - Present

Omni-Reason

Project Lead and First Author; research in progress

- Led video-model post-training, designing data mixtures and experiments and analyzing model behavior; adjusted task sampling, learning rates, and training duration to produce a single model capable of performing **100 computer vision tasks**.
- Built an agent-assisted data-production platform integrating automated data construction, human quality review, and GitHub PR tracking; coordinated 20+ research contributors to construct approximately **30M input-target video pairs across 350 tasks** for model training.

Training Object Permanence in World Models

Preprint, 2026

- Refactored 100 team-authored, single-scene Blender scripts into parameterized generators, enabling scalable data production for the project's 1.5M-sample object-permanence training corpus.

VBVR Research / Video-Reason Dec 2025 - Present

A Very Big Video Reasoning Suite (VBVR)

ICML 2026

- Led the design of all visual reasoning tasks across perception, transformation, spatial reasoning, abstraction, and knowledge, translating cognitive abilities into programmable tasks with explicit scenes, instructions, and target outcomes.
- Generated and rendered data for the full task suite, building a video-reasoning resource with 1M training examples for model training and systematic evaluation.

VBVR-Pro: A Scalable and Verifiable Suite for Native Visual Reasoning

Preprint, 2026

- Designed 150 new visual reasoning tasks beyond the original VBVR suite, expanding the combined coverage to 300 tasks.

Demystifying Video Reasoning

ECCV 2026

- Analyzed layer-wise representations across 40 DiT blocks, identifying spatially sparse token activations and a shift from global structure toward task-relevant objects.

Grow AI Like a Child Feb 2025 - Nov 2025

Egocentric Bias in Vision-Language Models (FlipSet)

CogSci 2026

- Led study design, evaluation of 103 vision-language models, analysis, and manuscript writing; developed controlled tasks to assess visual perspective-taking.
- Designed controls separating perspective judgment from mental rotation, revealing systematic egocentric bias and failures to compose component abilities.

Assessing Perceptual Metacognition in Vision-Language Models

CogSci 2026

- Scaled numerosity, orientation, and brightness tasks into experimental stimuli; evaluated models and analyzed confidence, calibration, and metacognitive measures to compare implicit confidence with explicit self-reports.

Vision Language Models Cannot Reason About Physical Transformation

ICML 2026

- Designed and filmed conservation and non-conservation tasks across number, length, size, and volume; computed and visualized performance correlations to investigate physical-transformation reasoning and model biases.